

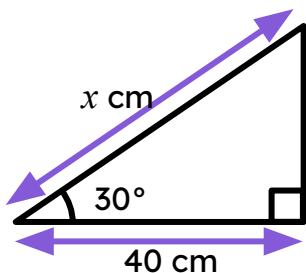


Choosing an appropriate method for finding lengths of a triangle

Task A: Trigonometry or not

1) Calculate x in each triangle.

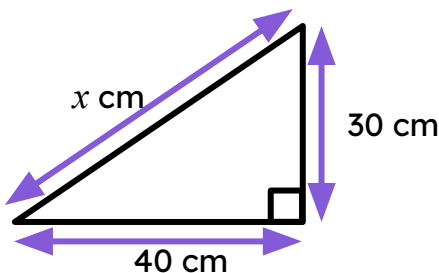
a)



$$x = \frac{40}{\cos(30^\circ)}$$

$$x = 46.2 \text{ (1 d.p.)}$$

b)

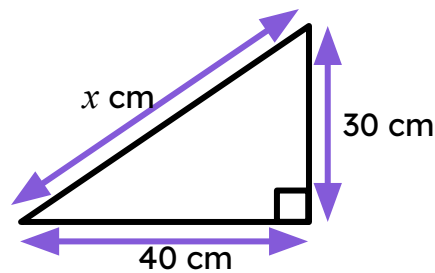


$$x^2 = 40^2 + 30^2$$

$$x^2 = 2500$$

$$x = 50$$

c) The perimeter is 1.2 m



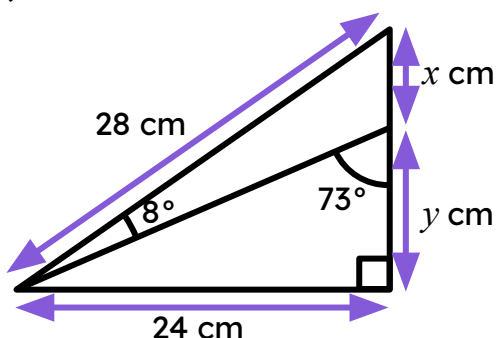
$$x + 40 + 30 = 120$$

$$x + 70 = 120$$

$$x = 50$$

2) Calculate x in each triangle, giving your answers to 1 decimal place.

a)



$$y = \frac{24}{\tan(73^\circ)}$$

$$y = 7.33...$$

$$28^2 = 24^2 + (x + 7.33...) ^2$$

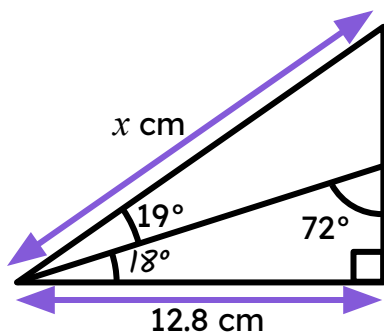
$$28^2 - 24^2 = (x + 7.33...) ^2$$

$$208 = (x + 7.33...) ^2$$

$$14.42... = x + 7.33...$$

$$7.1 = x \text{ (1 d.p.)}$$

b)

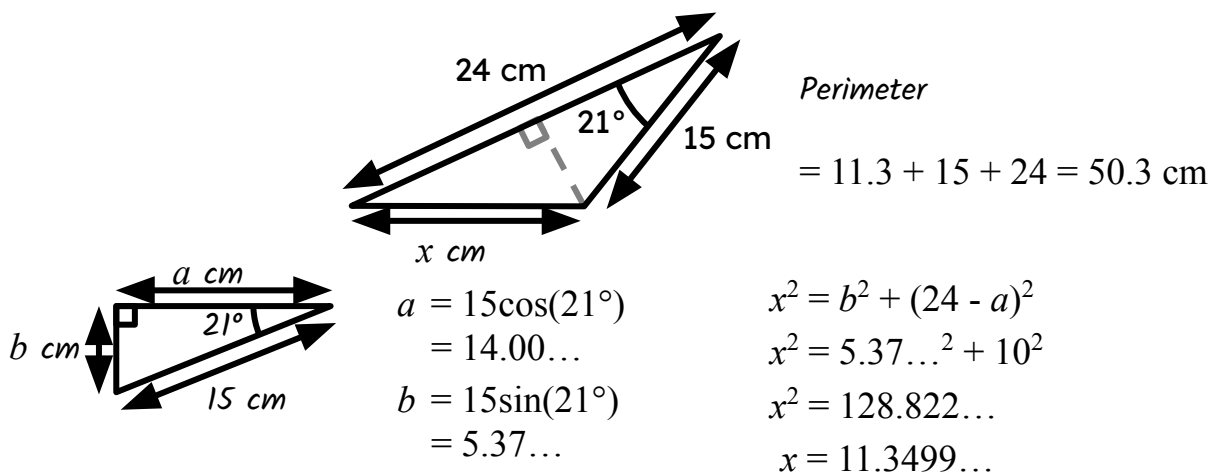


$$x = \frac{12.8}{\cos(37^\circ)}$$

$$x = 16.0$$



3) Work out the perimeter of this triangle, giving your answer to 1 decimal place.



Task B: Practical uses of trigonometry

1) A ramp is being built outside a house.

The bottom of the door is 42 cm off the floor.

The angle between the ramp and the ground should not exceed 4.8°

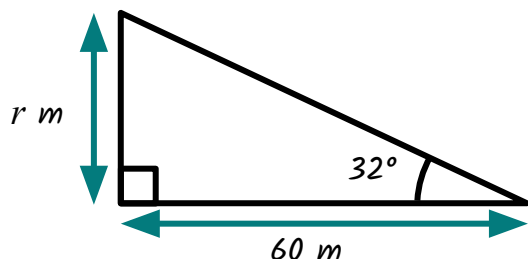
What is the minimum slope length of the ramp?



$$m = \frac{42}{\sin(4.8^\circ)} = 501.9$$

The slope is a minimum of 501.9 cm or 5.02 m long.

- 2) A man standing on a river-bank notes a particular tree on the opposite bank. He then walks 60 metres along the river-bank in a straight line. On looking back he can see that the angle between the line of his path and a line to the tree is 32° . How wide is the river?

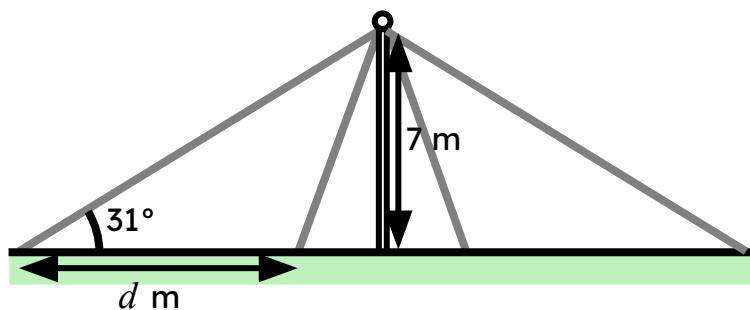


$$r = 60 \times \tan(32^\circ)$$

$$r = 37.5$$

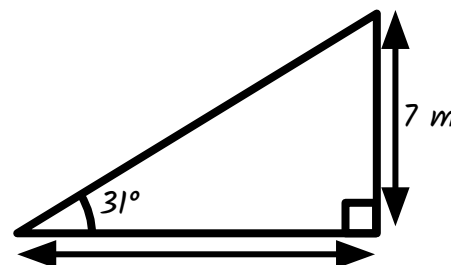
The river is 37.5 metres wide.

- 3) A 7 metre pole is secured by support wires. The wires are pegged in a straight line. The outer wires make an angle of 31° with the floor and the inner wires make an angle twice the size. How far apart are the wires?

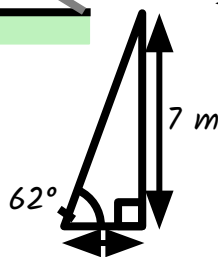


$$d = 11.65 - 3.72 = 7.93$$

The wires are 7.93 m apart.



$$\frac{7}{\tan(31^\circ)} = 11.65 \text{ m}$$



$$\frac{7}{\tan(62^\circ)} = 3.72 \text{ m}$$

